MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-B VOC (HONOURS) REGULAR EXAMINATION NOVEMBER 2025
FIRST SEMESTER
MINOR PATHWAY COURSES (MPC) - MG1MPCESM100 ENVIRONMENT AND ECOLOGY

Part A
Short Answer
Answer any 15 questions
Each question carries 1 mark

1. **List the scope of ecology. [K] [CO2]**
   To understand the structure and function of nature
   To maintain a healthy planet for the survey of life on earth
   To emphasize man's dependence on nature
   To recognize man's role as a responsible steward
   To enrich other branches of science
   To apply ecological principles in the welfare of man
   For the conservation and management of natural resources
   To promote sustainable development
   [Write any of the four points]
2. **Demonstrate a grazing Food chain.[U] [CO4]**
   Grass → Grasshopper → Frog → Snake → Hawk;
3. **Examine Clumped population distribution.[An] [CO5]**
   In this type of distribution the individuals of a species are grouped in forms of patches or clumps.
4. **Define habitat niche. [K] [CO3]**
   The physical space occupied by an organism.
5. **List the four realms of natural environment. [K] [CO1]**
   Lithosphere, hydrosphere, atmosphere, and biosphere.
6. **Discuss the biotic components of the environment. [U] [CO1]**
   Producers, consumers and decomposers.
7. **Describe mangrove ecosystem.[U] [CO3]**
   A mangrove ecosystem is a unique coastal ecosystem found in tropical and subtropical regions, where salt-tolerant trees and shrubs called mangroves grow along shorelines, estuaries, and river mouths**.**
8. **Identify the two types of nutrients. [U] [CO3]**
   Micronutrients and Macronutrients**.**
9. **Discuss cold desert. [U] [CO3]**
   A **cold desert** is a dry region where the temperature remains very low, rainfall is very little (mostly snow).
10. **Describe about the biotic components of forest ecosystem.[U] [CO3]**
    Biotic components of a forest ecosystem include all living organisms such as trees, shrubs, herbs, animals, birds, insects, microorganisms like fungi and bacteria.
11. **Indicate examples for lotic ecosystem.[U] [CO3]**
    River, streams and springs
12. **List one example for ecosystem.[K] [CO2]**

Forest ecosystem, Grassland ecosystem, Desert ecosystem, River ecosystem, Mangrove ecosystem, Wetland ecosystem etc

13. **Interpret shallow ecology.[U] [CO2]**

Shallow ecology focuses on protecting the environment mainly for human benefit rather than for the intrinsic value of nature.

14. **Identify the characteristic features of estuarine ecosystem. .[U] [CO3]**

Estuarine ecosystems are characterized by the mixing of freshwater and seawater, resulting in high nutrient levels and rich biodiversity.

15. **Discuss the importance of wetland ecosystem. .[U] [CO3]**

Wetland ecosystems are important because they purify water, control floods, support rich biodiversity, and act as crucial habitats for many species.

16. **Discuss dominant species. [U] [CO3]**

Dominant species are those that are most abundant or have the greatest influence on the structure and functioning of an ecosystem.

17. **Describe energy flow in an ecosystem. .[U] [CO4]**

Energy flow in an ecosystem is the one-way transfer of energy from producers to consumers and then to decomposers through food chains.

Part B

Short Essay

Answer any 3 questions

Each question carries 5 marks

18. **Differentiate between autecology and synecology. [U] [CO2]**

Autecology: It is the subdivision of ecology which deals with the ecological study of one species of organism throughout its life in relation to the ecological habitat factors like the life history, population dynamics, behaviour, home range etc. thereby concentrating on the finer details of a particular organism. It sharply focuses on a particular organism with a purpose of seeing how it fits into general ecological picture.

For example: Relationship of an oak tree to its environment.

Synecology: It deals with the ecological studies of communities or entire ecosystems, describing the overall energy and material flow through the system. Hence it is the study of groups of organisms in relation to its environment which are associated together as a unit.

For example: tropical forests or deserts.

19. **Identify the characteristics of grassland ecosystem. [U] [CO3]**

Grassland ecosystems are characterized by the dominance of grasses and herbaceous vegetation, with trees being absent or sparsely distributed due to moderate and seasonal rainfall. These ecosystems typically occur in regions receiving 25–75 cm of annual precipitation, which is insufficient for forest formation but adequate for sustaining extensive grass cover. Grasslands support diverse faunal communities, particularly large grazing herbivores such as bison, antelope, zebras, and deer, along with carnivores like lions, cheetahs, wolves, and various scavengers. Burrowing animals, insects, reptiles, and ground-nesting birds are also abundant. The ecological interactions within grasslands contribute to nutrient cycling, energy flow, and ecosystem stability.

20. **Distinguish between specific and ecological mortality. [U] [CO5]**

Natality refers to the average number of offspring produced by a population per unit time. The tendency to multiply is an inherent ability of a population. There are two types of natality:

i) Maximum (Absolute or Potential or Physiological) natality:

- It is theoretical maximum production of new individuals under ideal or optimal conditions. Usually it is a constant for a given population.
- This is never attained in nature owing to several adverse causes.

- For example, the maximum natality of a fish (Salmon) is 28000000 eggs in a season. But all the eggs do not hatch and develop into adults because of unfavorable environmental conditions.

ii) **Ecological or realized natality**: It refers to normal birth rate under actual or specific environmental conditions.

- It is not a constant for a population but may vary with the size and fecundity of the population and the physical environmental conditions.
- Ecological natality is always lesser than the potential natality.

21. Discuss major types of ecosystems. [U] [CO6]

Ecosystems are broadly classified based on their physical environment, climatic conditions, and the nature of their biotic and abiotic components. The major types of ecosystems include **terrestrial ecosystems**, and **aquatic ecosystems**.

1. Terrestrial Ecosystems: These ecosystems occur on land and are primarily influenced by climate, soil type, and vegetation. Major terrestrial ecosystems include:

Forest Ecosystems: Characterized by dense tree cover, high biodiversity, and complex food webs. Examples: tropical rainforests, temperate forests, boreal forests.

Grassland Ecosystems: Dominated by grasses with few trees; fertile soils and large grazing animals are common.

Desert Ecosystems: Very low rainfall, extreme temperatures, sparse vegetation, and drought-adapted flora and fauna.

Tundra Ecosystems: Found in Polar Regions with low temperatures, permafrost, and limited vegetation like mosses and lichens.

2. Aquatic Ecosystems

These ecosystems occur in water and are categorized into freshwater and marine types.

Freshwater Ecosystems: include Rivers, lakes, ponds, streams, and wetlands. They support diverse species adapted to low salinity.

Marine Ecosystems: Cover the largest part of the Earth's surface, including oceans, coral reefs, estuaries, and coastal zones. High salinity and great biological diversity characterize marine systems.

22. Differentiate between species richness and evenness. [U] [CO6]

Species diversity refers to the variety of species within a region. The species diversity of a region is measured on the basis of two parameters: Species richness and equitability or evenness

Species Richness: refers to the number of species per unit area. The number of species increases increases with the area of the site. Generally, higher species diversity represents greater species diversity.

Species Evenness or equitability: indicates the evenness in the number of individuals of a species. The evenness of species represents higher species diversity in the region.

Part C
Essay
Answer any 2 questions
Each question carries 10 marks

23. Describe components of ecosystem. [U] [CO3]

An ecosystem comprises two very closely interconnected parts, called Components. These are the living parts called biotic components and the non-living parts called abiotic components. Biotic component is an important part of ecosystem in which energy flows in the form of food from one organism to another organism within a community. Abiotic components are another part of ecosystem comprising inorganic, organic, and physical or climatic components. Together, these two components determine the characteristics of the ecosystem.

(1) Biotic Components

Biotic components are basically divided into following categories-

(i) Producers or Autotrophs

(ii) Consumers or Heterotrophs

(a) Primary consumers - herbivores

(b) Secondary consumers - carnivores

(c) Tertiary or top consumers carnivores

(iii) Parasites and Detrivores

(iv) Decomposers

Producers are green plants or autotrophs which synthesize their own food by the process called photosynthesis.

Consumers are heterotrophic organisms which directly or indirectly depend on producers for energy. Energy is procured in the form of food. Consumers have been divided into three categories-

(a) Primary consumers. These organisms are herbivores which depend upon green plants for their food, e.g. cow, goat, rabbit, some insects etc.

(b) Secondary consumers. These are those carnivores that use herbivores as their source of food, e.g. snake, frog etc.

(c) Tertiary consumers. These are also called top consumers and these kill and eat the secondary or primary consumers. Example of top carnivores are lion, tiger, hawk etc.

Parasites are those organisms, which obtain their food directly from living organisms (hosts). **Detrivores** or scavengers are small animals, which feed on dead bodies of other organisms, such as termites.

Decomposers are generally saprophytic microbes which decompose the dead organisms, such as bacteria and fungi. These decompose organic complexes into simple substances, which are

either mixed within the soil or evaporated to the atmosphere. These are also called micro consumers.

Producers, consumers and decomposers remain linked together by food chain wherein each category forms a trophic level.

ABIOTIC COMPONENTS

The abiotic or non-living components include the climatic and edaphic factors. The climatic factors include temperature, humidity, rain and snowfall etc. The edaphic factors comprise the soil and substratum. The ability of organisms to utilize, tolerate or combat the various abiotic factors are different and it may limit their distribution, behavior and relationship with other organisms. The most important abiotic factors are given below:

1. **Temperature:** Living organisms can survive only in a narrow range of temperature which allows their metabolism. The living organisms develop physiological and behavioral adaptations to withstand extremes of temperature. Many animal (such as some birds and mammals) migrate to warmer places in winter to avoid extreme cold. Some desert animals live inside burrows to avoid the intense heat of the desert.
2. **Water:** Water in an essential requirement of life. No life can exist without water. The requirement of water varies from organism to organism. The distribution of organisms depends upon the extent of the need and special adaptations for conserving water.
3. **Light:** Light is essential for photosynthetic organisms for the preparation of food on which the rest of the living world depends. Plants show various adaptations for obtaining optimum light. The activities of a large number of organisms are regulated by light. For example, cockroaches, moths and bats are active during the night.
4. **Humidity**: It is the amount of water vapours present in the atmosphere. Humidity regulates the rate at which water evaporates from the body surface of land organisms by transpiration, perspiration and other means. Different plants and animals show various adaptations to withstand dry conditions.
5. **Wind:** Wind determines the weather condition. It helps in the dispersal of seeds and fruits of many plants.
6. **Mineral elements**: A large number of minerals are required by organisms for their proper growth. Deficiency or absence of any one results in abnormal growth which may lead to death. High concentration of minerals are equally harmful and can limit the distribution of organisms.

24. Explain positive interspecific interactions among species. [U] [CO5]

Species interactions form the basis for many ecosystem properties and processes such as nutrient cycling and food webs. The interactions among the individuals of the same species is known as intraspecific interaction and that among the individuals of different species is called interspecific interaction. These relationships between species may be beneficial to both, harmful to both or harmful to one and neutral for the other.

<u>Positive Interactions:</u>

Positive interactions play a critical but underappreciated role in ecological communities by reducing physical or biotic stresses in existing habitats and by creating new habitats on which many species depend.

1. Commensalism

2. Facultative mutualism
3. Obligate mutualism
   (Explain with examples)

25. Identify the concept of ecotone and edge effect.

The zone where two major communities meet and blend together is known as an ecotone or a tension zone. In other words, an ecotone is a transition zone between two or more communities. The extent of ecotones is variable. The environmental conditions in the ecotone are intermediate between the two neighboring communities. An ectone presents a situation of special ecological interest. The border between forest and grassland, the transition between pond and dry land, the estuarine and the bank of a stream running through a meadow are some good examples of ecotone.

A remarkable feature of ectone is that it contains a great number of species and a higher population density than either of the neighboring communities. This tendency of ectone is known as edge effect or principle of edges. The species, which occur primarily or abundantly in the ecotone, are often called edge species.

For e.g. the **mangrove forests** represent an ecotone between marine and terrestrial ecosystem.

Other examples are **grassland** (between forest and desert), **estuary** (between fresh water and salt water) and **riverbank or marshland** (between dry and wet).

Features of Ecotones

- A sharp vegetation transition for example change in grass colours indicates an ecotone.
- Physiognomy: a key indicator of ecotone where the plant species differ in physical appearance.
- Change in species is indicators of ecotone where we observe some specific organisms at one side of the ecotone boundary while some other specific organism on the other side.
- Spatial mass effect: New plant establishment or migration obscures an ecotone as they cannot form self-sustaining population in different ecotone. But if survives between two communities, then species richness is exhibited by the ecotone**.**
- An ecotone can reveal the space sharing efficiency of two communities and the types of biomes by observing the exotic species abundance in ecotone.
- Best model to study diverse ecosystem.
- Shift in dominance represented by ecotone.
- Ecotone act as an ecological niche for the species colonizing at the junction called edge effect.

26. Explain population ecology. [U] [CO6]

Population ecology is a branch of ecology that studies the dynamics of species populations and how these populations interact with the environment. It focuses on understanding the factors that influence population size, density, distribution, growth, and structure over time. By analyzing these patterns, population ecology helps explain how species survive, reproduce, and

respond to ecological pressures, forming the basis for wildlife management, conservation biology, and environmental planning.

One of the fundamental concepts in population ecology is **population size**, which refers to the total number of individuals in a given area. Population density, the number of individuals per unit area or volume, provides insights into the intensity of interactions among individuals, such as competition for resources or breeding opportunities. **Population distribution** describes how individuals are spaced within their habitat—whether uniformly, randomly, or in clumps. Clumped distribution is the most common, occurring when resources are unevenly distributed or social behaviors promote grouping.

Population ecology also examines **age structure**, the proportion of individuals in different age groups within a population. Age structure influences growth potential because younger individuals contribute more to future reproduction. Closely connected to this is **natality (birth rate)** and **mortality (death rate)**, two demographic factors that directly determine population growth. **Immigration and emigration**, the movement of individuals in and out of populations, also play crucial roles in shaping population dynamics.

Population growth follows distinct patterns. Under ideal conditions—with abundant resources and no limiting factors—populations grow exponentially. This is represented by the J-shaped curve, indicating rapid and unchecked increase. However, such growth is rarely sustained in natural environments. As resources become limited, competition intensifies, leading to logistic growth, described by an S-shaped curve. The logistic model incorporates the concept of **carrying capacity (K)**, the maximum population size that the environment can support over time. When population size nears the carrying capacity, growth slows and eventually stabilizes.

Environmental factors that regulate population size are broadly categorized into **density-dependent** and **density-independent** factors. Density-dependent factors, such as disease, predation, and competition, have stronger effects as population density increases. Density-independent factors, including natural disasters and climatic conditions, affect populations regardless of size. These regulatory mechanisms ensure that populations do not grow indefinitely and help maintain ecological balance.